Approximate Policy Iteration and Variants

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Review 1: $Q^\star/V^\star$

$V^\star(s) :=$ maximum expected total reward starting from state s

$Q^\star(s,a) :=$ maximum expected total reward starting from state s and taking action a **for one step**, and then following the optimal strategy

Value Iteration to approximate $Q^\star/V^\star$: **(for finding the optimal policy)**

For $i = 1, 2, \ldots$

$$Q_i(s,a) \leftarrow R(s,a) + \gamma \sum_{s'} P(s'|s,a)\, V_{i-1}(s') \quad \text{for all } (s,a)$$

$$V_i(s) = \max_a Q_i(s,a) \quad \text{for all } s$$

Review 2: Q^π / V^π

Fix a policy π

$V^\pi(s) :=$ expected total reward starting from state s and following policy π

$Q^\pi(s,a) :=$ expected total reward starting from state s and taking action a for one step, and then following policy π

Approximate Q^π / V^π : **(for evaluating a given policy)**

For $i = 1, 2, \ldots$

$$Q_i(s,a) \leftarrow R(s,a) + \gamma \sum_{s'} P(s'|s,a)\, V_{i-1}(s') \quad \text{for all } (s,a)$$

$$V_i(s) = \sum_a \pi(a|s)\, Q_i(s,a) \quad \text{for all } s$$

Policy Iteration

Another way (other than value iteration) to find the optimal policy in an MDP

Policy Iteration

Policy Iteration

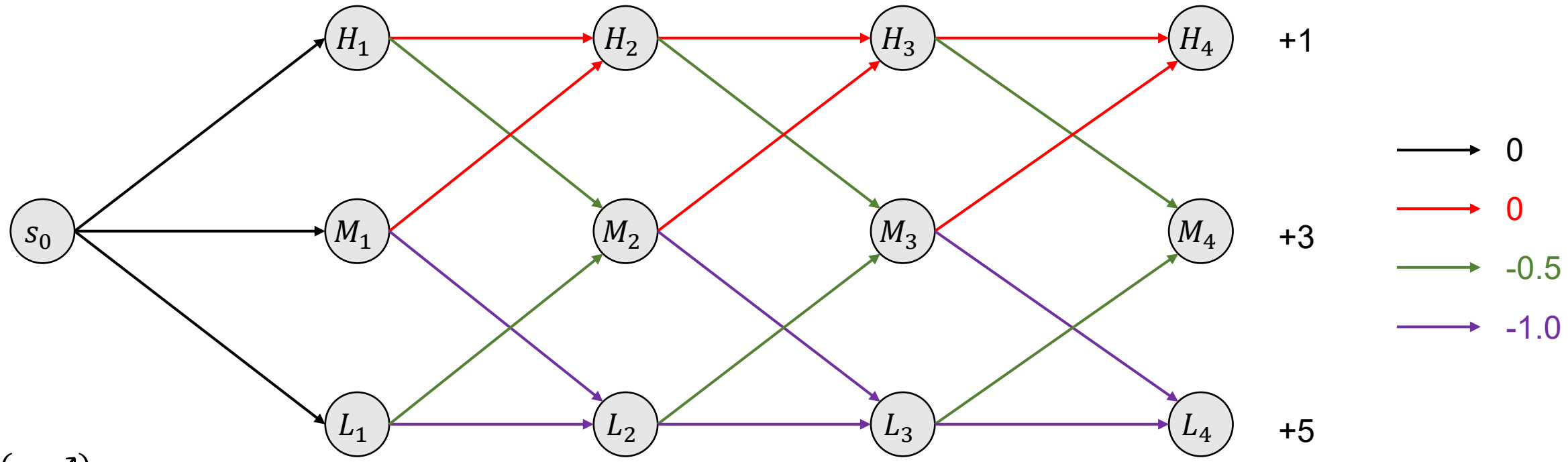
For $i = 1,\ 2, \ldots$

$$\forall s, \qquad \pi_i(s) \leftarrow \operatorname*{argmax}_a Q^{\pi_{i-1}}(s, a)$$

Theorem (monotonic improvement). Policy Iteration ensures

$$\forall s, a, \qquad Q^{\pi_i}(s, a) \geq Q^{\pi_{i-1}}(s, a)$$

When converged (i.e., $\pi_i = \pi_{i-1}$), we have $\pi_i = \pi^\star$.

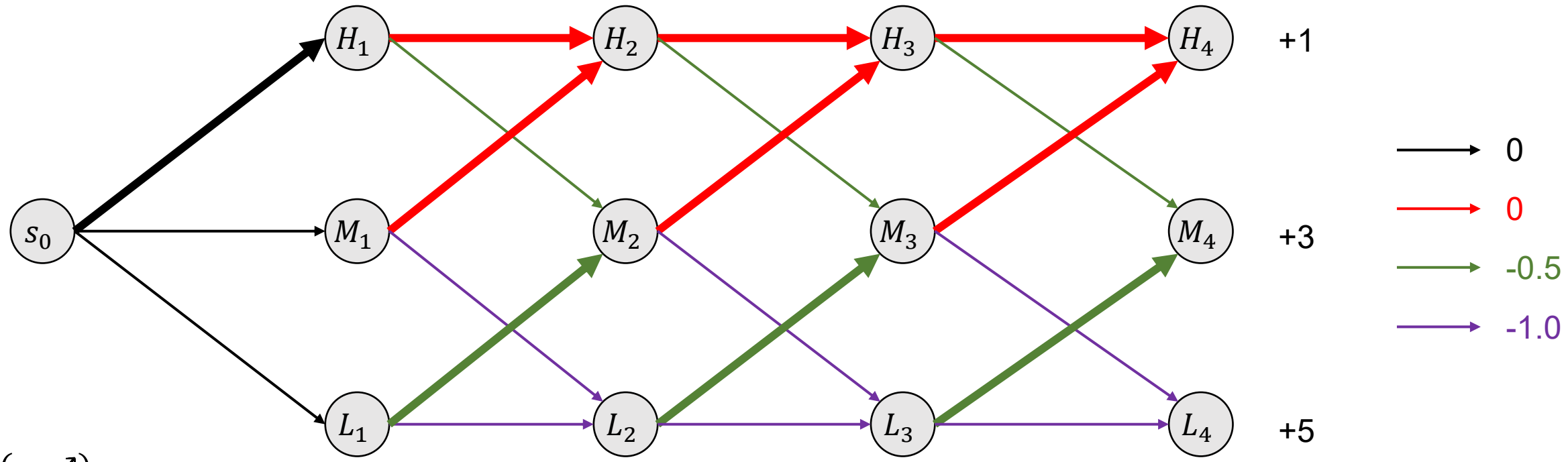


$Q^\pi(s_0, \nearrow) =$

$Q^\pi(s_0, \rightarrow) =$

$Q^\pi(s_0, \searrow) =$

$Q^\pi(H_1, R) =$	$Q^\pi(H_2, R) =$	$Q^\pi(H_3, R) =$
$Q^\pi(H_1, G) =$	$Q^\pi(H_2, G) =$	$Q^\pi(H_3, G) =$
$Q^\pi(M_1, R) =$	$Q^\pi(M_2, R) =$	$Q^\pi(M_3, R) =$
$Q^\pi(M_1, P) =$	$Q^\pi(M_2, P) =$	$Q^\pi(M_3, P) =$
$Q^\pi(L_1, G) =$	$Q^\pi(L_2, G) =$	$Q^\pi(L_3, G) =$
$Q^\pi(L_1, P) =$	$Q^\pi(L_2, P) =$	$Q^\pi(L_3, P) =$



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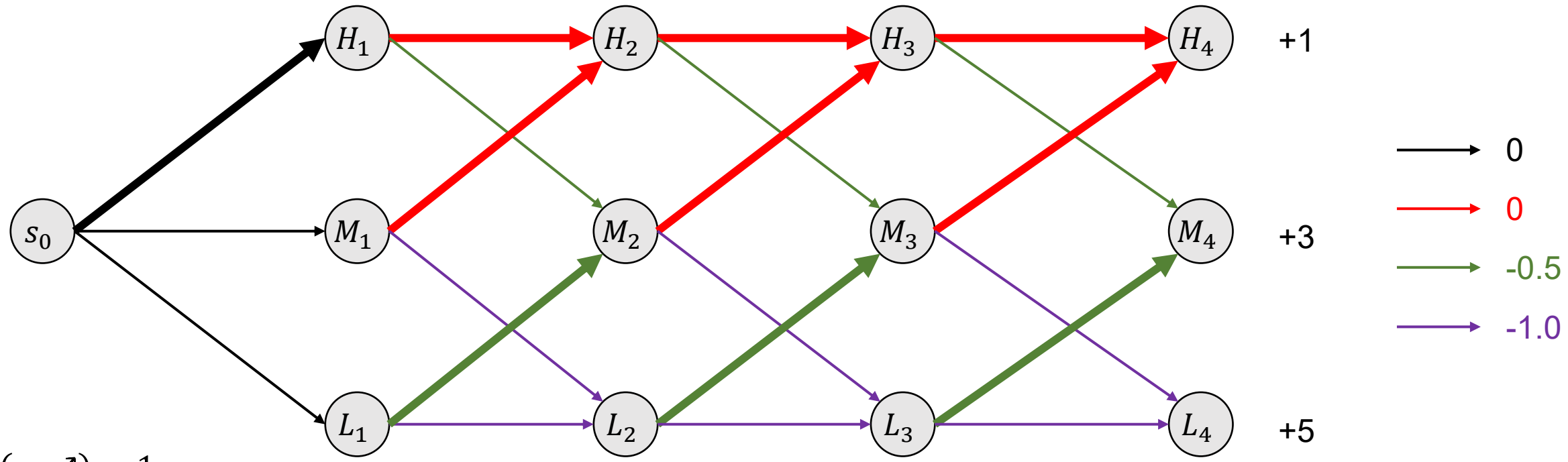
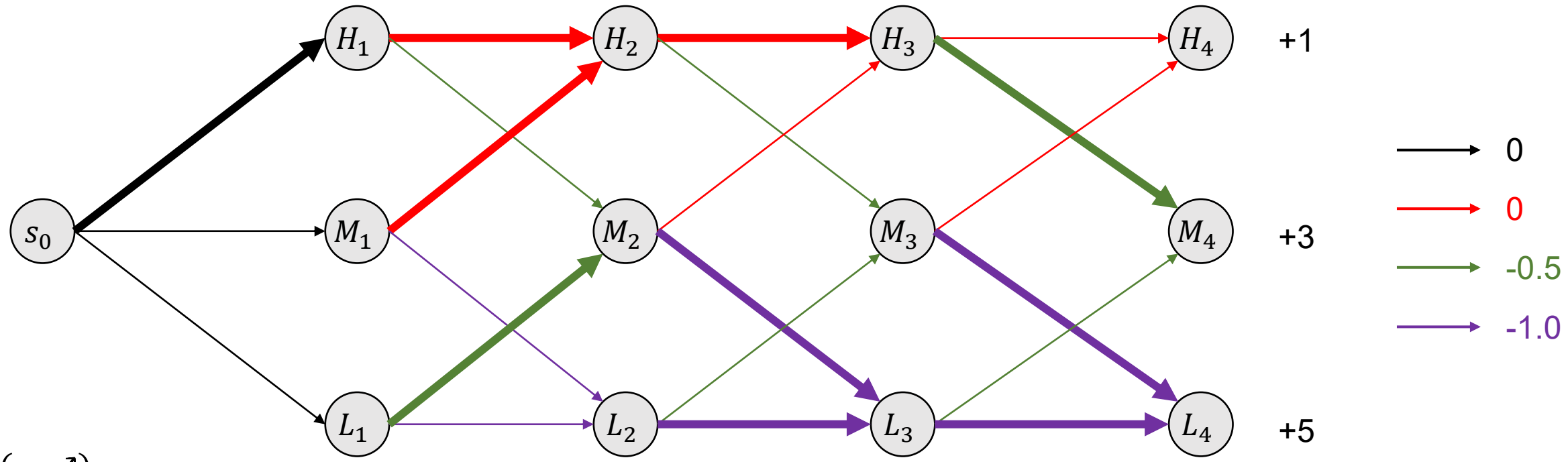


$Q^\pi(s_0, \nearrow) = 1$

$Q^\pi(s_0, \rightarrow) = 1$

$Q^\pi(s_0, \searrow) = 0.5$

$Q^\pi(H_1, R) = 1$	$Q^\pi(H_2, R) = 1$	$Q^\pi(H_3, R) = 1$
$Q^\pi(H_1, G) = 0.5$	$Q^\pi(H_2, G) = 0.5$	$Q^\pi(H_3, G) = 2.5$
$Q^\pi(M_1, R) = 1$	$Q^\pi(M_2, R) = 1$	$Q^\pi(M_3, R) = 1$
$Q^\pi(M_1, P) = -0.5$	$Q^\pi(M_2, P) = 1.5$	$Q^\pi(M_3, P) = 4$
$Q^\pi(L_1, G) = 0.5$	$Q^\pi(L_2, G) = 0.5$	$Q^\pi(L_3, G) = 2.5$
$Q^\pi(L_1, P) = -0.5$	$Q^\pi(L_2, P) = 1.5$	$Q^\pi(L_3, P) = 4$



$Q^\pi(s_0, \nearrow) =$

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$Q^\pi(H_1, R) =$

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$Q^\pi(M_1, P) =$

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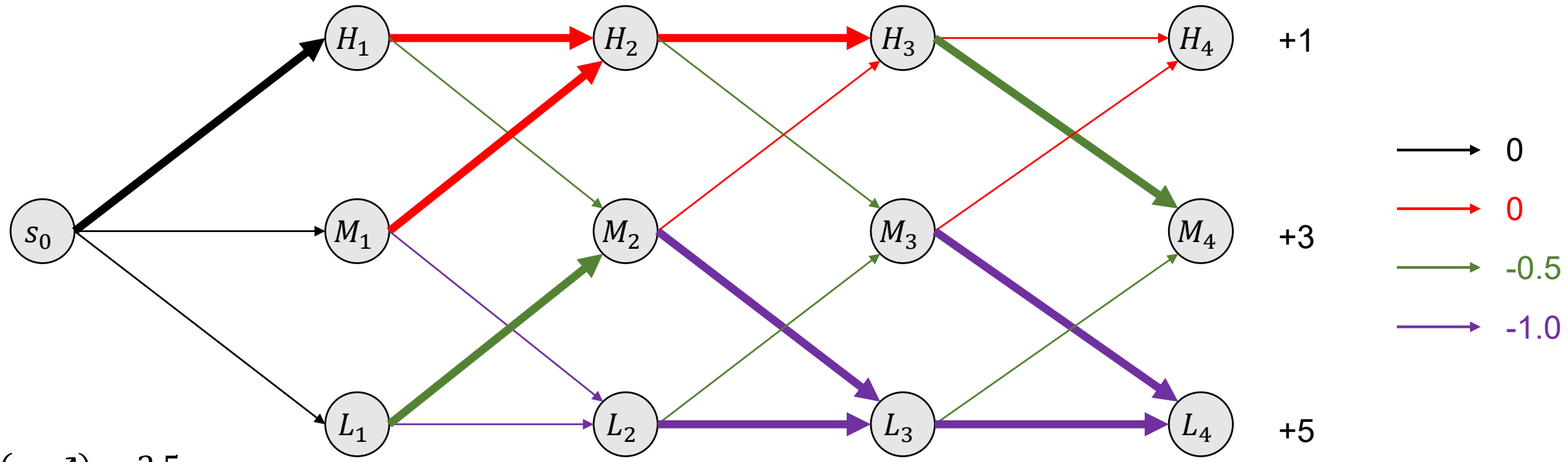

$Q^\pi(s_0, \nearrow) = 2.5$

$Q^\pi(s_0, \rightarrow) = 2.5$

$Q^\pi(s_0, \searrow) = 2.5$

$Q^\pi(H_1, R) = 2.5$	$Q^\pi(H_2, R) = 2.5$	$Q^\pi(H_3, R) = 1$
$Q^\pi(H_1, G) = 2.5$	$Q^\pi(H_2, G) = 3.5$	$Q^\pi(H_3, G) = 2.5$
$Q^\pi(M_1, R) = 2.5$	$Q^\pi(M_2, R) = 2.5$	$Q^\pi(M_3, R) = 1$
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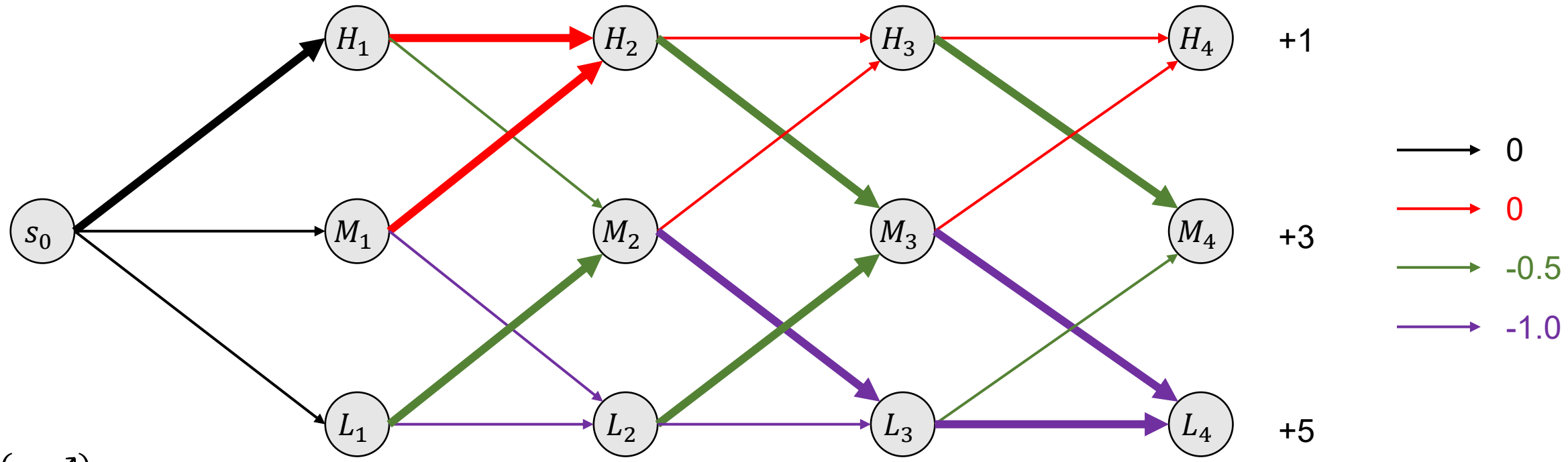
$Q^\pi(L_1, P) = 2$	$Q^\pi(L_2, P) = 3$	$Q^\pi(L_3, P) = 4$



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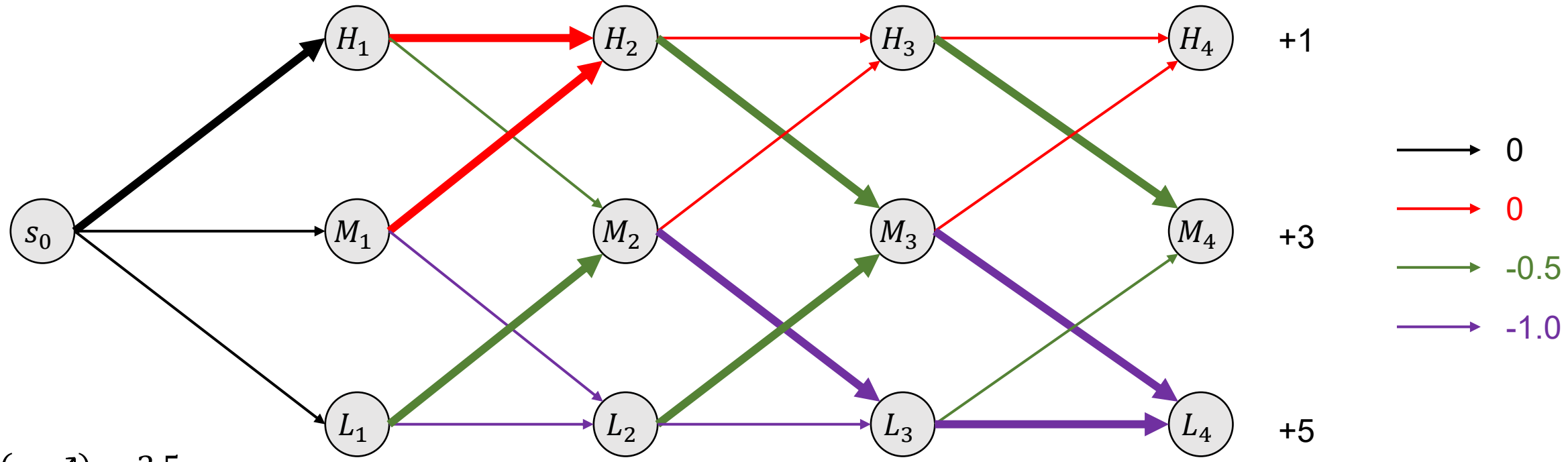

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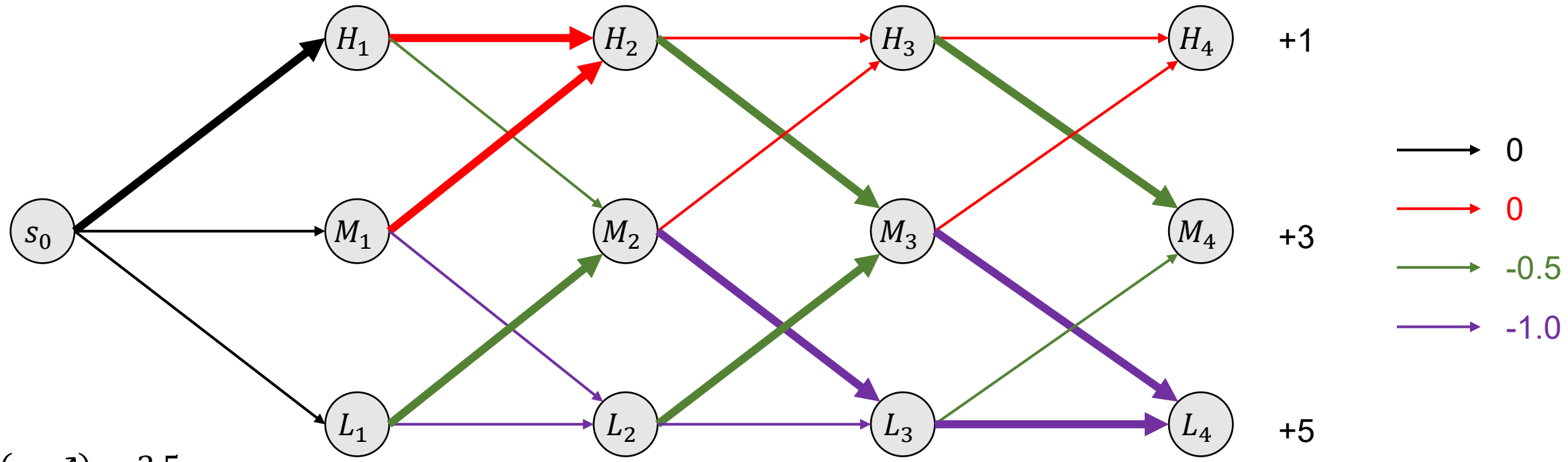

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$Q^\pi(H_1, G) = 2.5$ $\quad Q^\pi(H_2, G) = 3.5$ $\quad Q^\pi(H_3, G) = 2.5$

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$Q^\pi(L_1, P) = 2.5$ $\quad Q^\pi(L_2, P) = 3$ $\quad Q^\pi(L_3, P) = 4$

$\pi_{i+1} = \pi_i \Rightarrow$ converged

Policy Iteration

For $k = 1,\ 2, \ldots$

Calculate $Q^{\pi_k}(s, a) \quad \forall s, a$

$\pi_{k+1}(s) = \underset{a}{\operatorname{argmax}}\, Q^{\pi_k}(s, a) \quad \forall s$

Policy Iteration with Samples

For $k = 1,\ 2, \ldots$

For $i = 1, 2, \ldots, N$:

Choose action $a_i \sim \pi_{\theta_k}(\cdot\,|s_i)$

Receive reward $r_i \sim R(s_i, a_i)$ and $s_i' \sim P(\cdot\,|s_i, a_i)$

$s_{i+1} = s_i'$ if episode continues, $s_{i+1} \sim \rho$ if episode ends

Data collection

Evaluate $Z_k(s,a) \approx Q^{\pi_{\theta_k}}(s,a)$ for $s = s_1, \ldots, s_N$ and all a

or $Z_k(s,a) \approx Q^{\pi_{\theta_k}}(s,a) - b_k(s)$for $s = s_1, \ldots, s_N$ and all a

Policy Evaluation

Update θ_{k+1} from θ_k using the estimators $\{Z_k(s_i, a)\}_{i=1}^N$

Using any technique we introduced for policy-based contextual bandits

Policy Improvement

Policy Evaluation with Data

Policy Evaluation

Given: a policy π

Evaluate $V^{\pi}(s)$ or $Q^{\pi}(s, a)$

Monte Carlo Estimation

Start from $(s_1, a_1) = (\hat{s}, \hat{a})$ and
execute policy π until the episode ends and obtain trajectory

$$s_1 = \hat{s}, a_1 = \hat{a}, r_1, s_2, a_2, r_2, \dots, s_\tau, a_\tau, r_\tau$$

Let $G = \sum_{h=1}^{\tau} \gamma^{h-1} r_h$

G is a natural estimator for $Q^\pi(\hat{s}, \hat{a})$

Policy Iteration with Samples

For $k = 1,\ 2, \ldots$

For $i = 1, 2, \ldots, N$:

Choose action $a_i \sim \pi_{\theta_k}(\cdot\,|s_i)$

Receive reward $r_i \sim R(s_i, a_i)$ and $s_i' \sim P(\cdot\,|s_i, a_i)$

$s_{i+1} = s_i'$ if episode continues, $s_{i+1} \sim \rho$ if episode ends

Data collection

Evaluate $Z_k(s,a) \approx Q^{\pi_{\theta_k}}(s,a)$ for $s = s_1, \ldots, s_N$ and all a

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Policy Evaluation

Update θ_{k+1} from θ_k using the estimators $\{Z_k(s_i, a)\}_{i=1}^N$

Using any technique we introduced for policy-based contextual bandits

Policy Improvement

Value Iteration for V^π / Q^π

Input: π

For $k = 1,\ 2, \ldots$

$$\forall s, \qquad V_k(s) \leftarrow \sum_a \pi(a|s) \left(R(s,a) + \gamma \sum_{s'} P(s'|s,a)\, V_{k-1}(s') \right)$$

Input: π

For $k = 1,\ 2, \ldots$

$$\forall s, a, \qquad Q_k(s,a) \leftarrow R(s,a) + \gamma \sum_{s',a'} P(s'|s,a)\, \pi(a'|s') Q_{k-1}(s',a')$$

On-Policy Policy Evaluation

Temporal Difference (TD) Learning for V^{π}

For $k = 1,\ 2, \ldots$

Collect $\{(s_i, a_i, r_i, s_i')\}_{i=1}^{N}$ using policy π

$$\theta_k \leftarrow \theta_{k-1} - \alpha \nabla_{\theta} \frac{1}{N} \sum_{i=1}^{N} \left(V_{\theta}(s_i) - r_i - \gamma V_{\theta_{k-1}}(s_i') \right)^2 \Bigg|_{\theta = \theta_{k-1}}$$

No target network needed because this is an **on-policy** problem.

This algorithm is also called TD(0)

Temporal Difference (TD) Learning for Q^{π}

For $k = 1,\ 2, \ldots$

Collect $\{(s_i, a_i, r_i, s_i')\}_{i=1}^{N}$ using policy π

$$\theta_k \leftarrow \theta_{k-1} - \alpha \nabla_{\theta} \frac{1}{N} \sum_{i=1}^{N} \left(Q_{\theta}(s_i, a_i) - r_i - \gamma \sum_{a} \pi(a|s_i') Q_{\theta_{k-1}}(s_i', a') \right)^2 \Bigg|_{\theta=\theta_{k-1}}$$

No target network needed because this is an on-policy problem.

Generalized Policy Iteration

$N = \infty \Rightarrow$ Policy Iteration

$N = 1 \Rightarrow$ Value Iteration for policy optimization

For $i = 1, 2, \ldots$

$$\pi_i(s) = \operatorname*{argmax}_a Q_i(s,a)$$ ← **Policy update**

$Q \leftarrow Q_i$

Repeat for N times:

$$Q(s,a) \leftarrow R(s,a) + \gamma \sum_{s',a'} P(s'|s,a)\, \pi_i(a'|s') Q(s',a')$$ ← **Value update**

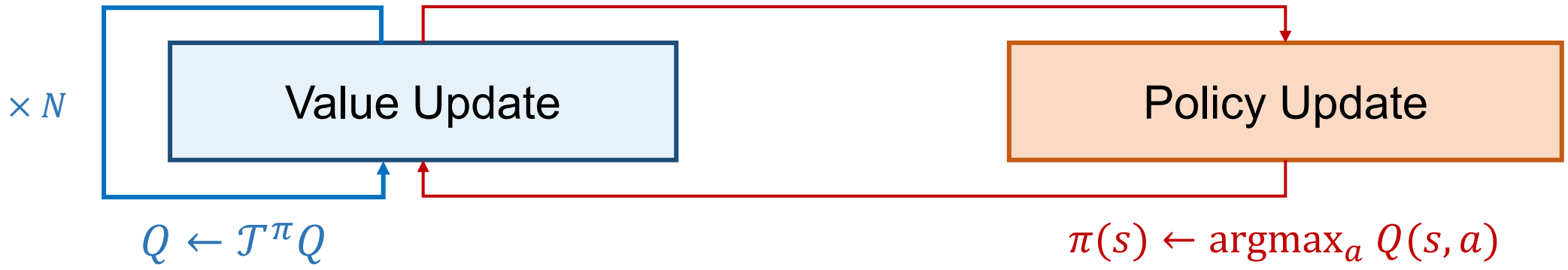
$Q_{i+1} \leftarrow Q$

Notice: in value iteration, there may not exist a policy π such that $Q_i = Q^\pi$

In contrast, in policy iteration we have $Q_i = Q^{\pi_{i-1}}$

VI for PO can be viewed as PI **with incomplete policy evaluation**

Generalized Policy Iteration



$Q \leftarrow \mathcal{T}^{\pi} Q$ means $Q(s,a) \leftarrow R(s,a) + \gamma \sum_{s',a'} P(s'|s,a)\, \pi(a'|s') Q(s',a')$ for all s, a

- Provide a unified view for algorithms approximating optimal policy with known P, R
- Provide a unified view for “value-based” and “policy-based” algorithms
- Compared with bandits:
  - The R is replaced by Q (could be Q^{π} or $Q^{\star}$) to capture the long-term goal of the learner

A Family of Estimators

Suppose we have a **state-value function estimation** $V_\phi(s) \approx V^\pi(s)$

Suppose we also have a **trajectory** $s_1, a_1, r_1, \ldots, s_\tau, a_\tau, r_\tau$ generated by π where $s_{\tau+1}$ is a terminal state

The following are all valid estimators of $Q^\pi(s_1, a_1)$:

$G_{1:1} = r_1 + \gamma V_\phi(s_2)$

…

$G_{1:\tau-1} = r_1 + \gamma r_2 + \gamma^2 r_3 + \cdots + \gamma^{\tau-1} V_\phi(s_\tau)$

$G_{1:\tau} = r_1 + \gamma r_2 + \gamma^2 r_3 + \cdots + \gamma^{\tau-1} r_\tau$

$G_{1:\tau+1} = r_1 + \gamma r_2 + \gamma^2 r_3 + \cdots + \gamma^{\tau-1} r_\tau$

…

A Family of Estimators

And the following are estimators of $Q^{\pi}(s_1, a_1) - V_{\phi}(s_1)$ (baseline)

$A_{1:1} = r_1 + \gamma V_{\phi}(s_2) - V_{\phi}(s_1)$

...

$A_{1:\tau-1} = r_1 + \gamma r_2 + \gamma^2 r_3 + \cdots + \gamma^{\tau-1} V_{\phi}(s_{\tau}) - V_{\phi}(s_1)$

$A_{1:\tau} = r_1 + \gamma r_2 + \gamma^2 r_3 + \cdots + \gamma^{\tau-1} r_{\tau} - V_{\phi}(s_1)$

$A_{1:\tau+1} = r_1 + \gamma r_2 + \gamma^2 r_3 + \cdots + \gamma^{\tau-1} r_{\tau} - V_{\phi}(s_1)$

…

Below, we will introduce a way to combine these estimators.

Balancing Bias and Variance

$$G_1(\lambda) = (1-\lambda)\sum_{i=1}^{\infty}\lambda^{i-1}G_{1:i}$$

$$= (1-\lambda)\left(G_{1:1} + \lambda G_{1:2} + \lambda^2 G_{1:3} + \cdots + \lambda^{\tau-1}G_{1:\tau} + \lambda^{\tau}G_{1:\tau+1} + \lambda^{\tau+1}G_{1:\tau+2} + \cdots\right)$$

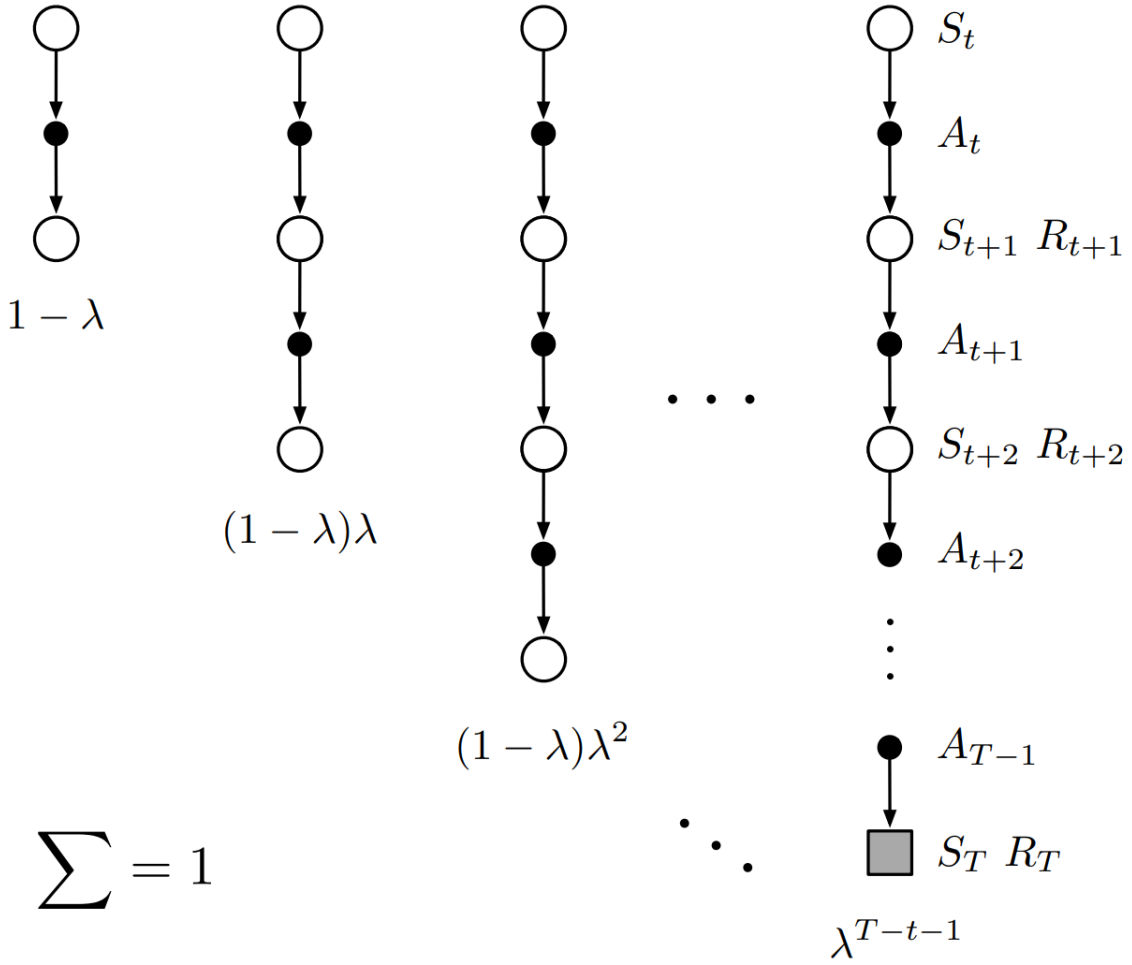
$$A_1(\lambda) = (1-\lambda)\sum_{i=1}^{\infty}\lambda^{i-1}A_{1:i}$$

$$= (1-\lambda)\left(A_{1:1} + \lambda A_{1:2} + \lambda^2 A_{1:3} + \cdots + \lambda^{\tau-1}A_{1:\tau} + \lambda^{\tau}A_{1:\tau+1} + \lambda^{\tau+1}A_{1:\tau+2} + \cdots\right)$$

$A_1(\lambda) = G_1(\lambda) - V_\phi(s_1)$ **(Generalized Advantage Estimation)**

An estimation of $Q^\pi(s_1, a_1) - V^\pi(s_1)$

Balancing Bias and Variance



Computing Generalized Advantage Estimator (GAE)

We also need to calculate

$$A_2(\lambda) = (1-\lambda)\sum_{i=2}^{\infty}\lambda^{i-2}A_{2:i} \quad = (1-\lambda)\left(A_{2:2} + \lambda A_{2:3} + \lambda^2 A_{2:4} + \cdots + \lambda^{\tau-2}A_{2:\tau} + \lambda^{\tau-1}A_{2:\tau+1} + \lambda^{\tau}A_{2:\tau+2} + \cdots\right)$$

$$A_3(\lambda) = (1-\lambda)\sum_{i=3}^{\infty}\lambda^{i-3}A_{3:i}$$

...

To compute $A_1(\lambda), A_2(\lambda) \ldots, A_\tau(\lambda)$, naïve implementation takes $\tau \times \tau \times \tau$ time.

Efficient Computation of GAE (1/2)

Define $\delta_i = r_i + \gamma V_\phi(s_{i+1}) - V_\phi(s_i)$

$$A_{i:j} = r_i + \gamma r_{i+1} + \gamma^2 r_{i+2} + \cdots + \gamma^{j-i} r_j + \gamma^{j-i+1} V_\phi\left(s_{j+1}\right) - V_\phi(s_i)$$

$$= \delta_i + \gamma \delta_{i+1} + \gamma^2 \delta_{i+1} + \cdots + \gamma^{j-i} \delta_j$$

Efficient Computation of GAE (2/2)

$$A_\tau(\lambda) = (1-\lambda)\left(A_{\tau:\tau} + \lambda A_{\tau:\tau+1} + \lambda^2 A_{\tau:\tau+2} + \cdots\right)$$

$$A_{\tau-1}(\lambda) = (1-\lambda)\left(A_{\tau-1:\tau-1} + \lambda A_{\tau-1:\tau} + \lambda^2 A_{\tau-1:\tau+1} + \cdots\right)$$

$$A_1(\lambda) = (1-\lambda)\left(A_{1:1} + \lambda A_{1:2} + \lambda^2 A_{1:3} + \cdots\right)$$

$$\boxed{A_i(\lambda) = \delta_i + \lambda\gamma A_{i+1}(\lambda)}$$

GAE (Generalized Advantage Estimation)

Let $(s_1, a_1, r_1, s_1', s_2, a_2, r_2, s_2', \dots, s_N, a_N, r_N, s_N')$ be a trajectory collected with policy π, where $s_i' = s_{i+1}$ if s_i' is not a terminal state, and $s_{i+1} \sim \rho$ otherwise.

Also, let V_ϕ be a given state-value estimation.

Then the following procedure can estimate $A_i \approx Q^\pi(s_i, a_i) - V_\phi(s_i)$

Parameter: λ (controlling variance-bias tradeoff)

For $i = N, N-1, \dots, 1$:

If s_i' is a terminal state:

$\delta_i = r_i - V_\phi(s_i)$

$A_i = \delta_i$

Else:

$\delta_i = r_i + \gamma V_\phi(s_{i+1}) - V_\phi(s_i)$

$A_i = \delta_i + \lambda\gamma A_{i+1}$

Schulman et al. High-Dimensional Continuous Control Using Generalized Advantage Estimation. 2015.

Using GAE in the Policy Iteration Framework

For $k = 1,\ 2, \ldots$

For $i = 1, 2, \ldots, N$:

Choose action $a_i \sim \pi_{\theta_k}(\cdot\,|s_i)$

Receive reward $r_i \sim R(s_i, a_i)$ and $s_i' \sim P(\cdot\,|s_i, a_i)$

$s_{i+1} = s_i'$ if episode continues, $s_{i+1} \sim \rho$ if episode ends

Data collection

Evaluate $Z_k(s, a) \approx Q^{\pi_{\theta_k}}(s, a) - V_\phi(s)$ for $s = s_1, \ldots, s_N$ and all a

$$\Rightarrow\ Z_k(s_i, a) = \frac{\mathbb{I}\{a_i = a\}}{\pi_{\theta_k}(a|s_i)}\,\hat{A}_k(s_i, a_i)$$

Policy Evaluation

Update θ_{k+1} from θ_k using the estimator $\{Z_k(s_i, a)\}_{i=1}^N$

Using any technique we introduced for policy-based contextual bandits

Policy Improvement

Training the Baseline V_ϕ (in iteration k)

For $k = 1, 2, \ldots$

$V_\phi(s) \approx V^{\pi_{\theta_k}}(s)$

For $i = 1, 2, \ldots, N$:

Choose action $a_i \sim \pi_{\theta_k}(\cdot \mid s_i)$

Receive reward $r_i \sim R(s_i, a_i)$ and $s_i' \sim P(\cdot \mid s_i, a_i)$

$s_{i+1} = s_i'$ if episode continues, $s_{i+1} \sim \rho$ if episode ends

$V^\pi(s)$

$r_i + \gamma V_{\phi_k}(s_{i+1})$

$r_i + \gamma r_{i+1} + \gamma^2 V_{\phi_k}(s_{i+2})$

$\vdots$

$$\phi_{k+1} \leftarrow \phi_k - \alpha \nabla_\phi \frac{1}{N} \sum_{i=1}^{N} \left(V_\phi(s_i) - r_i - \gamma V_{\phi_k}(s_i') \right)^2 \Big|_{\phi=\phi_k}$$ TD(0)

$$\phi_{k+1} \leftarrow \phi_k - \alpha \nabla_\phi \frac{1}{N} \sum_{i=1}^{N} \left(V_\phi(s_i) - G_i(\lambda; \phi_k) \right)^2 \Big|_{\phi=\phi_k}$$ where $G_i(\lambda; \phi_k) = A_i(\lambda; \phi_k) + V_{\phi_k}(s_i)$ TD(λ)

$$\phi_{k+1} \leftarrow \phi_k - \alpha \nabla_\phi \frac{1}{N} \sum_{i=1}^{N} \left(V_\phi(s_i) - \sum_{h=i}^{\tau(i)} \gamma^{h-i} r_i \right)^2 \Big|_{\phi=\phi_k}$$ TD(1)

Approximate Policy Iteration and Variants

PPO

GAE $A_i(\lambda)$

For $k = 1,\ 2, \ldots$

Requires training a separate V_ϕ

For $i = 1, 2, \ldots, N$:

Choose action $a_i \sim \pi_{\theta_k}(\cdot\,|s_i)$

Receive reward $r_i \sim R(s_i, a_i)$ and $s_i' \sim P(\cdot\,|s_i, a_i)$

$s_{i+1} = s_i'$ if episode continues, $s_{i+1} \sim \rho$ if episode ends

Define $Z_k(s_i, a) = \frac{\mathbb{I}\{a_i = a\}}{\pi_{\theta_k}(a|s_i)} \hat{A}_k(s_i, a_i)$

Use another inner for-loop to solve the argmax with gradient ascent

$$\theta_{k+1} = \underset{\theta}{\operatorname{argmax}} \left\{ \frac{1}{N} \sum_{i=1}^{N} \left(\sum_{a} \pi_\theta(a|s_i) Z_k(s_i, a) - \frac{1}{\eta} \mathrm{KL}\big(\pi_{\theta_k}(\cdot\,|s_i), \pi_\theta(\cdot\,|s_i)\big) \right) \right\}$$

$$\approx \underset{\theta}{\operatorname{argmax}} \left\{ \frac{1}{N} \sum_{i=1}^{N} \left(\frac{\pi_\theta(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)} \hat{A}_k(s_i, a_i) - \frac{1}{\eta} \left(\frac{\pi_\theta(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)} - 1 - \log \frac{\pi_\theta(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)} \right) \right) \right\}$$

Schulman et al. Proximal Policy Optimization Algorithms. 2017.

PPO with Clipping

Schulman et al. Proximal Policy Optimization Algorithms. 2017.

① Clipping ② KL-regularization

$$\theta_{k+1} = \underset{\theta}{\operatorname{argmax}} \left\{ \frac{1}{N} \sum_{i=1}^{N} \left(\boxed{} - \frac{1}{\eta} \left(\frac{\pi_\theta(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)} - 1 - \log \frac{\pi_\theta(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)} \right) \right) \right\}$$

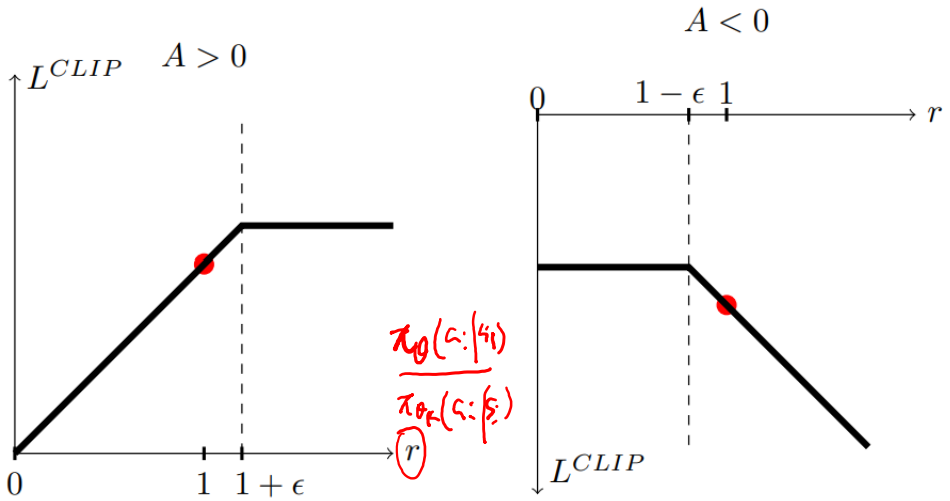
$< 1-\varepsilon$

$$L^{CLIP} \triangleq \boxed{\min \left\{ \frac{\pi_\theta(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)} \hat{A}_k(s_i, a_i), \quad \text{clip}_{[1-\epsilon, 1+\epsilon]} \left(\frac{\pi_\theta(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)} \right) \hat{A}_k(s_i, a_i) \right\}}$$

smaller

$1-\varepsilon$

$\epsilon \simeq 0.1, 0.2$



$\frac{\pi_\theta(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)}$

Preventing $\frac{\pi_{\theta_{k+1}}(a_i|s_i)}{\pi_{\theta_k}(a_i|s_i)} \hat{A}_k(s_i, a_i)$ from being too high

~~from being too low~~

A2C (Advantage Actor Critic) / PG

For $k = 1,\ 2, \ldots$

For $i = 1, 2, \ldots, N$:

Choose action $a_i \sim \pi_{\theta_k}(\cdot\,|s_i)$

Receive reward $r_i \sim R(s_i, a_i)$ and $s_i' \sim P(\cdot\,|s_i, a_i)$

$s_{i+1} = s_i'$ if episode continues, $s_{i+1} \sim \rho$ if episode ends

$$\theta_{k+1} = \theta_k - \eta \frac{1}{N} \sum_{i=1}^{N} \left(\nabla_\theta \log \pi_\theta(a_i|s_i)\right)\Big|_{\theta=\theta_k} \hat{A}_k(s_i, a_i)$$

In standard A2C, $\hat{A}_k(s_i, a_i) = \underline{r_i + \gamma V_{\phi_k}(s_i') - V_{\phi_k}(s_i)}$ (GAE estimator with $\lambda = 0$)

and ϕ_k is trained with TD(0):

$$\phi_{k+1} \leftarrow \phi_k - \alpha \nabla_\phi \frac{1}{N} \sum_{i=1}^{N} \left(V_\phi(s_i) - r_1 - \gamma V_{\phi_k}(s_i')\right)^2 \Bigg|_{\phi=\phi_k}$$

A2C (Advantage Actor Critic) / PG

For $k = 1,\ 2, \ldots$

For $i = 1, 2, \ldots, N$:

Choose action $a_i \sim \pi_{\theta_k}(\cdot\,|s_i)$

Receive reward $r_i \sim R(s_i, a_i)$ and $s_i' \sim P(\cdot\,|s_i, a_i)$

$s_{i+1} = s_i'$ if episode continues, $s_{i+1} \sim \rho$ if episode ends

$$\theta_{k+1} = \theta_k - \eta \frac{1}{N} \sum_{i=1}^{N} \left(\nabla_\theta \log \pi_\theta(a_i|s_i)\right)\Big|_{\theta=\theta_k} \hat{A}_k(s_i, a_i)$$

In standard PG, $\hat{A}_k(s_i, a_i) = \boxed{\sum_{h=i}^{\tau(i)} \gamma^{h-i} r_i} - V_{\phi_k}(s_i)$ (GAE estimator with $\lambda = 1$)

A2C (Advantage Actor Critic) / PG

For $k = 1,\ 2, \ldots$

For $i = 1, 2, \ldots, N$:

Choose action $a_i \sim \pi_{\theta_k}(\cdot\,|s_i)$

Receive reward $r_i \sim R(s_i, a_i)$ and $s_i' \sim P(\cdot\,|s_i, a_i)$

$s_{i+1} = s_i'$ if episode continues, $s_{i+1} \sim \rho$ if episode ends

$$\theta_{k+1} = \theta_k - \eta \frac{1}{N} \sum_{i=1}^{N} \left(\nabla_\theta \log \pi_\theta(a_i|s_i)\right)\Big|_{\theta=\theta_k} \hat{A}_k(s_i, a_i)$$

V_ϕ, λ

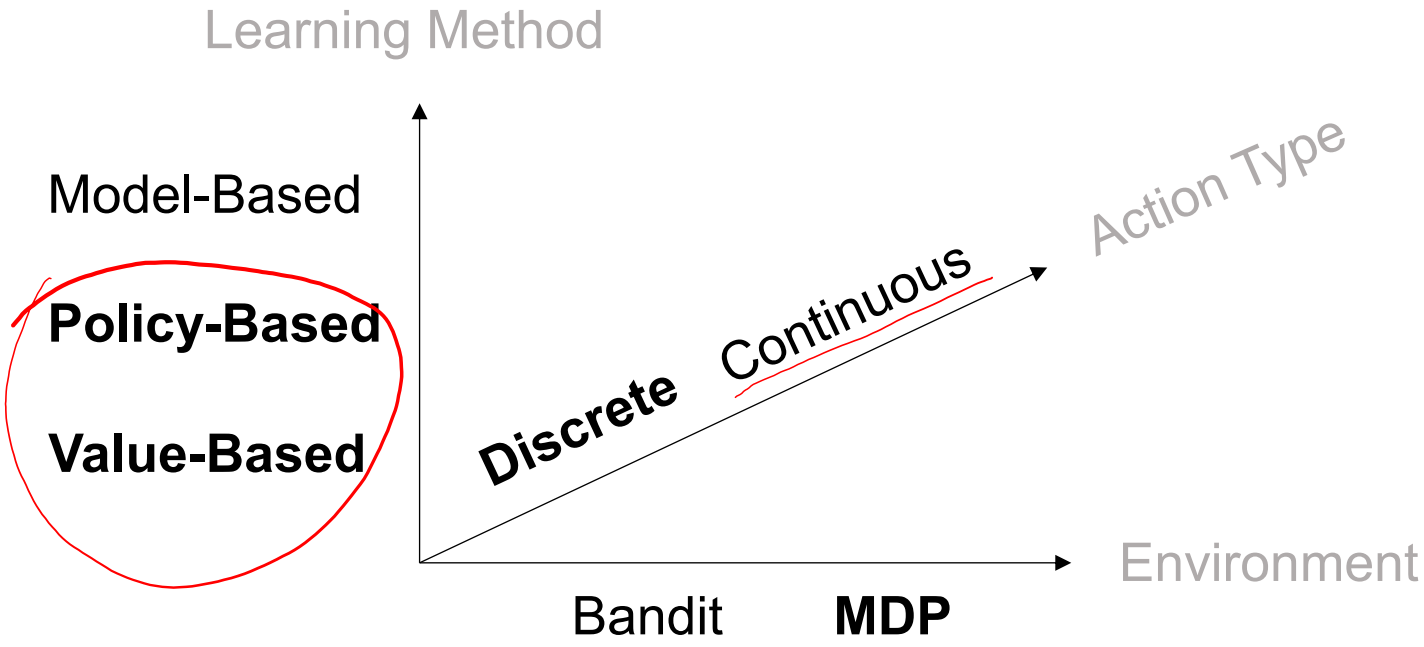
In general, one can use GAE with any λ to calculate $\hat{A}_k(s_i, a_i)$, with V_ϕ calculated from TD(λ') with any λ'.

Summary: Algorithms based on Policy Iteration

- The algorithms are almost the same as those we introduced for contextual bandits
  - PPO / NPG ← KL regularized / Clipped
  - A2C / PG ←
- The only change is replacing $r(x_i, a_i) - b(x_i)$ by Advantage Estimator:
  - $\lambda = 0$: $r(s_i, a_i) + \gamma V_\phi(s_{i+1}) - V_\phi(s_i)$
  - $\lambda = 1$: $r(s_i, a_i) + \gamma r(s_{i+1}, a_{i+1}) + \gamma^2 r(s_{i+2}, a_{i+2}) + \cdots + \gamma^{\tau - i}\, r(s_\tau, a_\tau) - V_\phi(s_i)$
  - Any $\lambda \in [0,1]$: calculated by the GAE procedure
- The baseline $V_\phi(s)$ tries to track $V^{\pi_\theta}(s)$ where π_θ is the current policy
  - It is trained with a separate procedure TD(λ')

$$\phi_{k+1} \leftarrow \phi_k - \alpha \nabla_\phi \frac{1}{N} \sum_{i=1}^{N} \left(V_\phi(s_i) - r_1 - \gamma V_{\phi_k}(s_i') \right)^2 \Bigg|_{\phi = \phi_k} \qquad \text{TD(0)}$$

Roadmap


Learning Method
Model-Based
Policy-Based
Value-Based
Action Type
Discrete
Continuous
Environment
Bandit
MDP